

- Intermediate DSA: Time & Space Complexity
- Intermediate DSA: Time Complexity
- Intermediate DSA: Introduction to Arrays & Basics of AI
- Intermediate DSA: Lab Session on TC, SC, Output & Debugging
- Intermediate DSA: Arrays: Prefix Sum & Carry Forward with AI-Assisted Problem Solving
- Intermediate DSA: Lab Session on Prefix Sum & Carry Forward
- Intermediate DSA: Arrays: Arrays techniques & smart prompting
- Intermediate DSA: Lab Session on Memory Management & Sorting Basics
- Intermediate DSA: Bit Manipulations Basics
- Intermediate DSA: 2D Matrices
- Intermediate DSA : Strings & Immutability visualisations through AI
- Intermediate DSA: Lab on Matrix & Strings using AI
- Introductory DSA Contest

$$\text{actor}(n) = x, \text{ then } n \% x == 0$$

Ques 1 : factors of the number 24.



24



1	x	24
2	x	12
3	x	8
4	x	6
6	x	4
8	x	3
12	x	2
24	x	1

Quiz 2

10
↓
1 x 10
2 x 5
5 x 2
10 x 1

~~7~~ 2

```
int countFactors (int n) {
```

```
    int ans = 0;
```

```
    for (int i = 1; i <= n; i++) {
```

```
        if (n % i == 0)
```

```
            ans ++;
```

```
    }
```

```
    return ans;
```

```
}
```

$$\Rightarrow 10^8 \Rightarrow 1 \text{ sec}$$

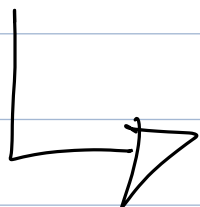
n

Time

$$10^8$$

$$1 \text{ sec}$$

$$10^{10} = \underline{\underline{100 \text{ seconds}}}$$



$$10^8 \text{ iterations} = \underline{1} \text{ second}$$

$$1 \text{ iteration} \Rightarrow \frac{1}{10^8} \text{ second}$$

$$10^{10} \text{ iterations} \Rightarrow \frac{1}{10^8} \times 10^{10}$$

$$\Rightarrow \frac{10^{10}}{10^8} \left| \frac{a^n}{a^m} \right| = a^{n-}$$

$$\Rightarrow 10^2$$

n

10^{18}

Time

10^{10} seconds.



317 years



24



1	x	24
2	x	12
3	x	8
4	x	6

6	x	4
8	x	3
12	x	2
24	x	1

$$i \times j = n$$

$$i < j$$

$$i > j$$

$$i \times j = n$$

$$j = \frac{n}{i}$$

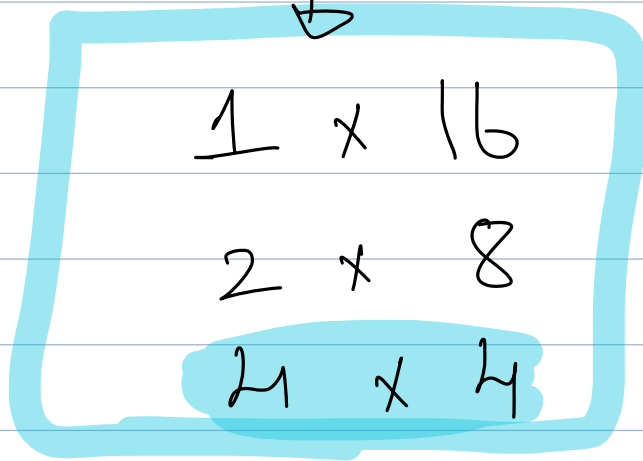
$$i < j$$

$$i < \frac{n}{i}$$

$$i^2 < n$$

$$i < \sqrt{n}$$

$$16 \Rightarrow 5$$



$$8 \times 2$$

$$16 \times 1$$

```
int count_factors (int n) {
```

```
    int ans = 0
```

```
    for (int i = 1; i*i <= n; i++) {
```

```
        if (n % i == 0) {
```

```
            if (i == n/i)
```

```
                ans = ans + 1
```

```
            else
```

```
                ans = ans + 2 ;
```

```
        }
```

```
    }
```

```
    return ans;
```

```
}
```

24

i	ans
<u>1</u>	2
2	4
3	6
4	8

ans = 8

16

n/i

i	ans
<u>1</u>	2
2	4
3	4
4	5

→ 5

n	Method 1	Method 2
<u>10^{18}</u>	$n = 10^{18}$ iterations = 10^{18} time = 10^{10} seconds ≈ 317 years.	$n = 10^{18}$ iterations = 10^9 time $\Rightarrow$ <u>10 seconds</u>

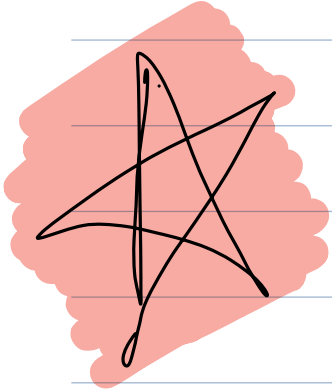
$$\sqrt{10^{18}} \Rightarrow \underline{\underline{10^9}}$$

x is a prime if it has only 2 factors (1 and x):

Quiz 10, 11, 23, 2, 25, 27, 31

$\rightarrow$ 4

$[3, 10]$ $\Rightarrow$ How many numbers
are in this range
both inclusive,



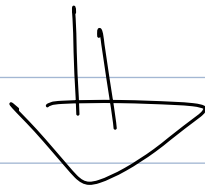
Range Counting.

$$[a, b] = b - a + 1$$

$$1 + 2 + 3 + \dots + 100$$

Natural numbers $\Rightarrow 1, 2, \dots, \infty$

$$\text{Sum of } 1^{\text{st}} \text{ } n \text{ natural numbers} \Rightarrow \frac{n(n+1)}{2}$$



$$\frac{100(101)}{2} \Rightarrow 5050$$

ToDo: Derive this formula !!

Geometric Progression

$$\Rightarrow 5, 10, 20, 40 \dots$$

$$a, ar, ar^2, ar^3 \dots ar^{n-1}$$

$$\Rightarrow 3, 9, 27, 81 \dots$$

$$a, ar, ar^2, ar^3 \dots ar^{n-1}$$

$$\text{Sum of } 1^{\text{st}} \text{ } n \text{ terms} \Rightarrow \frac{a(r^n - 1)}{r - 1}$$

Quiz 5

for ($i \rightarrow 1$ to n) {

if ($i == n$)
break:

}

$i = [1, n]$

$\rightarrow n - 1 + 1$

$\rightarrow n$

Quiz 6

for ($i \rightarrow 0$ to 100) {

$S = S + i + i^2$:

}

$i \rightarrow [0, 100] \rightarrow 100 - 0 + 1$

$\rightarrow \underline{101}$

How many iterations will be there in this loop?

```
func(){  
    for(i -> 1 to N){  
        if(i % 2 == 0){  
            print(i);  
        }  
    }  
    for(j -> 1 to M){  
        if(j % 2 == 0){  
            print(j);  
        }  
    }  
}
```

Iⁿ

I^m

$\Rightarrow \underline{\underline{n + m}}$

Algo 1
[Mukul]

10 seconds

↓ Same
machine

(Python) 2 seconds

2 seconds



Algo 2
[Saijan]

5 seconds

[better
machine]

5 seconds

↓ python.

1.5 seconds

Fair Judgement $\rightarrow$ No of iterations!!

Asymptotic Analysis

$\rightarrow$ How does an algorithm behave for large inputs.

Big O.

$\rightarrow$ Count the number of iterations.

$\rightarrow$ Drop the lower order terms

$\rightarrow$ Drop the constants

Ex 1

$$\text{iterations} = 4n^2 + 3n + 1$$

Drop
lower
Order
terms

$$= 4n^2 + 3n + 1$$

Drop
Constants

$$= 4n^2$$

$$\Rightarrow O(n^2)$$

$$\log n < \sqrt{n} < n < n \log n < n\sqrt{n} < n^2 \\ < n^3 < 2^n < n! < n^n$$

Quisg ~~$4n + 3n \log n + 1$~~

$$\Rightarrow O(n \log n)$$

Quiz 9 ~~$4n \log n + 3n\sqrt{n} + 10^6$~~

$$\Rightarrow O(n\sqrt{n})$$

→ Why do we neglect the lower order terms

Σn Total iteration = $n^2 + 10n$

n	Total ($n^2 + 10n$)	Contribution ($10n$)	%
10	200	100	50%
100	$10^4 + 10^3$ ≈ 11000	10^3 1000	9%
10^4	$10^8 + 10^5$	10^5	<u>0.1%</u>

Why do we ignore constants.

Algo 1	Algo 2	Winner
$10 \log n$	n	Algo 1
$10^4 \log n$	n	Algo 1

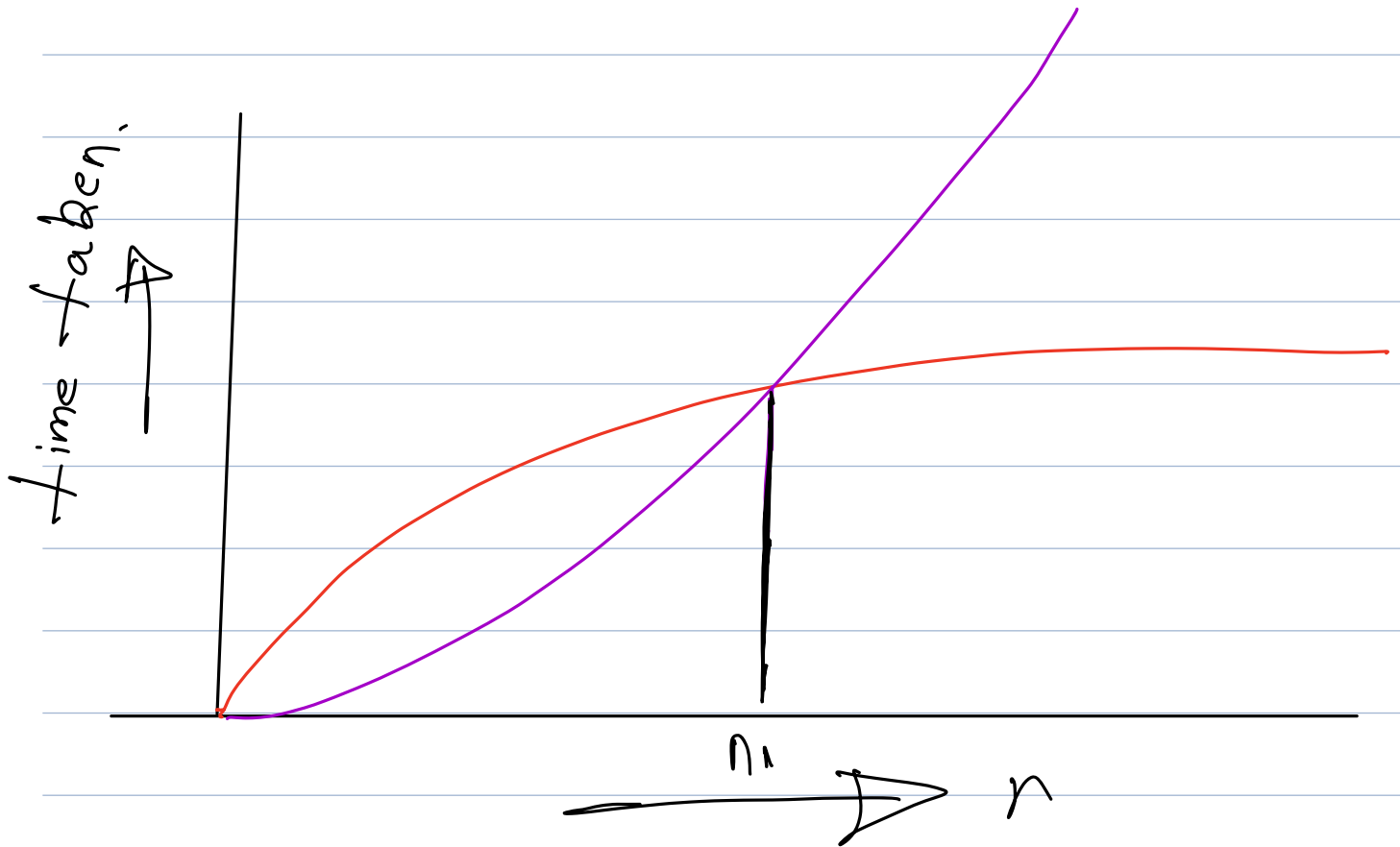
Issues with Big O.

<u>1st Issue</u> :	Algo 1	Algo 2
	$10^2 n$	n
	$\Downarrow$	$\Downarrow$
	$O(n)$	$O(n)$

2nd Issue

Algo 1

Algo 2



Before n_1 = purple is better.

After n_1 = red is better.